

NIKEL, Russian Federation

WMO: 220040

Lat: 69.410N

Lon: 30.209E

Elev: 302

StdP: 14.54

Time Zone: 3.00 (E03)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.4	-7.7	-18.9	2.0	-14.5	-12.4	2.8	-7.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	76.6	60.5	71.8	59.4	68.1	57.8	62.9	73.6	60.3	70.0	58.3	66.9	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.1	73.0	67.7	55.5	66.3	64.4	53.7	62.0	62.9	28.5	74.0	26.7	70.0	25.4	67.6	69.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-20.5	80.4	4.0	4.4	-23.3	83.6	-25.6	86.1	-27.9	88.6	-30.7	91.8		
(5)			-20.8	64.7	4.0	3.6	-23.7	67.2	-26.0	69.3	-28.2	71.4	-31.1	74.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.5	12.9	18.4	23.4	31.1	42.3	49.3	57.2	54.3	47.5	33.9	24.3	18.3
	DBStd	17.00	11.85	10.73	8.80	6.90	7.98	8.07	8.25	6.39	6.45	6.54	8.53	10.31
	HDD50	6218	1151	885	825	566	265	109	21	24	122	499	770	981
	HDD65	11172	1616	1305	1290	1016	704	477	269	340	525	964	1220	1446
	CDD50	562	0	0	0	0	27	87	245	156	47	0	0	0
	CDD65	41	0	0	0	0	1	5	28	7	0	0	0	0
	CDH74	244	0	0	0	0	14	17	182	31	0	0	0	0
	CDH80	44	0	0	0	0	1	2	33	8	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	17.8	1.3	0.9	0.7	0.6	0.9	2.0	2.6	2.5	2.0	1.5	1.4	1.4
	PrecMax	22.4	2.6	2.6	1.3	1.5	2.4	6.8	5.7	5.4	4.8	2.6	4.0	3.5
	PrecMin	11.9	0.1	0.1	0.1	0.1	0.2	0.4	0.5	0.9	0.5	0.4	0.2	0.0
	PrecStd	2.7	0.6	0.5	0.3	0.3	0.5	1.3	1.3	0.9	0.9	0.6	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	39.6	41.3	51.4	74.2	75.7	84.8	78.1	65.9	49.9	39.8	38.5
		MCWB	35.7	36.1	35.1	42.1	57.3	60.3	63.2	64.2	57.2	47.0	36.8	35.5
	2%	DB	37.0	36.4	38.5	45.5	64.0	70.8	79.0	71.4	61.4	46.9	37.7	36.2
		MCWB	34.0	34.0	33.7	39.2	50.3	58.0	61.5	61.0	53.5	44.4	35.6	33.8
	5%	DB	31.9	33.9	36.4	41.5	59.0	66.5	74.6	67.4	58.8	44.6	35.9	33.9
		MCWB	29.6	31.7	32.4	36.7	48.2	55.0	60.4	58.7	51.8	42.1	34.3	32.0
	10%	DB	28.8	31.7	34.2	38.8	54.3	62.2	70.3	63.9	56.2	42.6	34.5	31.4
		MCWB	27.0	29.7	31.0	34.9	45.9	53.3	58.3	56.9	50.4	40.5	32.8	29.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	36.7	35.8	43.2	56.5	62.9	66.9	66.0	57.6	47.8	37.7	35.9
		MCDB	38.8	39.1	40.0	50.2	71.7	73.2	74.2	75.9	65.4	48.9	39.1	37.8
	2%	WB	33.8	34.1	34.2	39.6	51.9	59.3	63.7	62.1	54.6	44.7	35.8	34.0
		MCDB	37.4	36.2	37.7	45.0	62.2	68.4	75.2	69.5	60.3	47.0	37.2	35.9
	5%	WB	29.7	32.0	32.8	36.9	48.5	56.4	61.3	59.4	52.3	42.6	34.2	32.1
		MCDB	31.8	33.6	35.9	40.3	57.9	64.9	73.2	66.5	57.5	44.4	35.9	34.0
	10%	WB	27.0	29.7	31.2	35.5	46.1	53.1	59.2	57.1	50.7	40.5	32.8	29.4
		MCDB	28.7	31.8	34.1	38.3	53.7	61.1	69.2	63.4	55.7	42.4	34.3	31.2
Mean Daily Temperature Range	5% DB	MDBR	9.7	8.0	8.3	8.2	9.9	9.9	11.2	8.8	7.4	5.1	7.0	9.4
		MCDBR	10.5	6.8	7.7	8.6	18.2	15.5	17.2	13.7	11.0	5.3	6.5	9.0
		MCWBR	9.5	6.2	5.8	5.7	9.9	7.4	6.9	6.7	6.3	4.9	6.9	8.4
	5% WB	MCDBR	10.3	7.0	7.3	8.0	17.2	15.1	14.9	12.9	9.7	5.3	6.9	8.7
		MCWBR	9.3	6.9	6.1	5.7	10.0	8.3	7.4	7.7	6.5	5.1	6.7	8.3
Clear-Sky Solar Irradiance	taub	0.207	0.253	0.301	0.331	0.356	0.352	0.367	0.349	0.313	0.272	0.361	N/A	
	taud	1.913	2.047	2.124	2.184	2.335	2.443	2.430	2.456	2.516	2.222	2.680	N/A	
	Ebn at Noon	7	151	221	253	262	268	258	249	229	146	18	0	
	Edh at Noon	1	14	25	33	32	30	29	25	18	11	3	0	
All-Sky Solar Radiation	RadAvg	4	117	543	1084	1427	1508	1391	954	514	152	17	0	
	RadStd	0	12	29	74	143	121	152	96	42	19	2	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page